

Machine Learning

Problem Set 01

Instructions. If you think any question is too difficult, you can give up strategically. Please arrange your time according to your own schedule.

Please make your best effort to submit the assignment as a PDF compiled using L^AT_EX. We do not accept handwritten and scanned documents and strongly discourage the use of Markdown. For other typesetting tools similar to L^AT_EX, the principle is to produce a document with a clear and professional layout comparable to that of L^AT_EX.

Regarding the use of AI, if you are unable to solve a problem and use an AI tool to get ideas, you must ensure that you fully understand its proposed solution. You are responsible for verifying the correctness of the answer and writing the solution yourself. This is especially important for proof-based questions. If the teaching assistants find a submitted answer suspicious, they may contact you to explain your problem-solving process.

If you received ideas from classmates, please acknowledge them in your submission. Appropriate discussions will not negatively affect your grade, but direct copying will result in a severe penalty.

These same requirements will apply to all future assignments and will not be repeated.

Question 1. Self-Supervised Learning and Challenges toward AGI

- (a) In addition to supervised, semi-supervised, and unsupervised learning, there is also a term called self-supervised learning. Please describe self-supervised learning and highlight how it differs from unsupervised learning.
- (b) Read *Unsupervised Representation Learning by Predicting Image Rotations*, and explain it using your own words.
- (c) To attain Artificial General Intelligence (AGI), identify and discuss one critical challenge that must be overcome. Specifically, please provide insights into the current capabilities and the envisioned future direction.

You may choose to explore challenges within various cutting-edge domains, such as advanced reasoning models, agentic reinforcement learning (RL), continual or lifelong learning, embodied AI (e.g., for robotics), computer vision, or the development of world models. You are encouraged to select any critical challenge you find relevant to achieving AGI.

(For example, it is a challenge to retrieve and update the world knowledge in the model instantly. While we humans can naturally update knowledge whenever we receive some new information, current models usually fix their parameters once trained. To solve this problem, there are many ways, including [memory-based methods](#), [additional parameters](#), and [fine-tuning](#). However, we still need to improve the efficiency, accuracy, and flexibility of knowledge editing to achieve AGI.)

Question 2. Chernoff bound

Given the Chernoff bound for bounded variables: Let $X_1, \dots, X_n$ be a sequence of i.i.d. random variables satisfying $0 \leq X_i \leq 1$. Let $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$. We have that, for $0 \leq \delta \leq 1$,

$$\Pr[X \geq (1 + \delta)\mu] \leq e^{-\delta^2\mu/3}.$$

(For materials related to the Chernoff bound, see lectures: [CSE 312](#), [COS 521](#), and [18.310](#).)

Now try to solve this problem using the Chernoff bound:

Consider a data center that receives an average of 6 server requests per minute, according to historical traffic data. The center becomes overloaded if it receives 10 or more requests in a minute. This data center has 15 independent servers, each with the same probability of receiving a request in any given minute (that is, a server will receive either one request or no request at all).

Please provide a bound for the probability that the data center will be overloaded.

Question 3. Smoothness and strong convexity

(a) Show the equivalence between the following two definitions of L -smoothness. (Assume f is twice continuously differentiable.)

(i) $\|\nabla f(w) - \nabla f(w')\| \leq L \|w - w'\|.$

(ii) $|f(w') - f(w) - \langle \nabla f(w), w' - w \rangle| \leq \frac{L}{2} \|w - w'\|^2.$

(b) All conditions below hold for all $x, y \in \mathbb{R}^n$ and $\alpha \in [0, 1]$. Let f be twice continuously differentiable. Show the equivalence among the following definitions of μ -strong convexity.

(i) $f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) - \alpha(1 - \alpha)\frac{\mu}{2} \|x - y\|^2.$

(ii) $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|x - y\|^2.$

(iii) $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \mu \|x - y\|^2.$

(iv) $\lambda_{\min}(\nabla^2 f(x)) \geq \mu.$

(Note: $\nabla^2 f(x)$ denotes the Hessian of the function $f(\cdot)$ at point x .)

Note. In Question 4, you can assume that f is twice continuously differentiable and try to use the conclusions from Question 3.

Question 4. Gradient descent

(a) If a function is L -smooth, but not convex, please show how Gradient Descent (with an appropriate learning rate) converges to a “stationary point,” which is defined as a point with zero gradient. The convergence rate should be given in terms of $\|\nabla f\|^2$ and the number of iterations T . We assume that the function to be optimized is lower bounded.

(b) If a function is L -smooth and μ -strongly convex, please show that Gradient Descent (with an appropriate learning rate) achieves linear convergence.